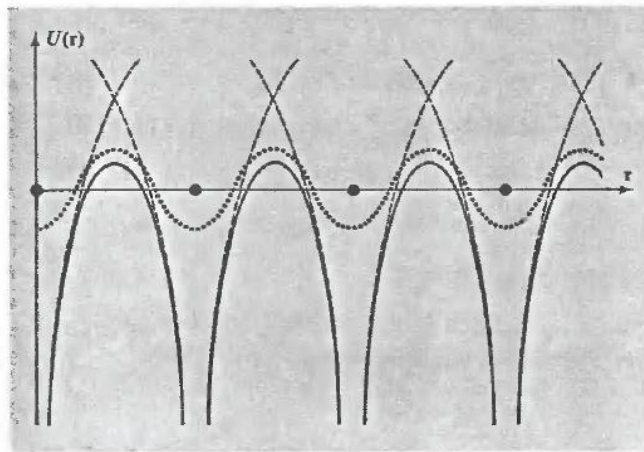


Electrons in a periodic potential:

Consider an infinitely long 1-dimensional chain of ions each separated by a distance “a”. The ions form a 1-d crystal with lattice parameter “a”. These ions give rise to an attractive potential (V) that is periodic in nature, i.e. $V(x + a) = V(x)$. This is shown schematically in the figure below.

**Figure 8.1**

A typical crystalline periodic potential, plotted along a line of ions and along a line midway between a plane of ions. (Closed circles are the equilibrium ion sites; the solid curves give the potential along the line of ions; the dotted curves give the potential along a line between planes of ions; the dashed curves give the potential of single isolated ions.)

We want to find out how an electron behaves under the influence of such a potential. For this we need to solve the Schrodinger equation which is:

$$\left(-\frac{1}{2}\nabla^2 + V\right)\psi = E\psi, \dots\dots\dots (1)$$

where the symbols having their usual meaning. From solid state physics you know that the wavefunction takes the form of Bloch function, i.e., $\psi_k(x) = e^{ikx}u_k(x)$, where $u_k(x)$ has the underlying periodicity of the Bravais lattice. k is a quantum number that is a consequence of periodic boundary condition (PBC) and lies between $-\frac{\pi}{a} \leq k < \frac{\pi}{a}$. The spacing between two k 's is given by $\Delta k = \frac{2\pi}{a} \cdot \frac{1}{N}$, where N is the number of unit cells in real space used in the PBC.

We will solve Eqn.1 using variational principle. To do this we will expand the periodic part of the Bloch function in a basis. Here we will be using a plane wave basis $\varphi_G = \frac{1}{\sqrt{a}}e^{iGx}$, where $G = \frac{2\pi}{a}m$, $m \in \text{all } I$ (integers) are the reciprocal lattice vectors. Since there are infinite G 's, so they form a complete basis. Under this expansion the Bloch function assumes the following form:

$\psi_k(x) = \frac{1}{\sqrt{a}} \sum_G c_k^G e^{i(k+G)x}$, where the c_k^G are the expansion coefficients and are to be determined.

To simplify things, let us consider a potential $V = -V_0 \left[1 + \cos\left(\frac{2\pi}{a}x\right)\right]$, $V_0 \geq 0$. For $V_0 = 0$, we will get the free electron model. For small values of V_0 , the potential acts as a perturbation

to the free electron (nearly free electron model). For both of these scenarios you can analytically solve this. For the second case you need to use perturbation theory. However, here we will solve them using numerical methods as asked below:

- (1) Workout the matrix elements corresponding to the kinetic and potential energy. (I will check this when grading. Needs to be done in pen and paper. No typed notes.)
- (2) Write a code (that will use V_0 , number of k-points, no. of G vectors as input parameters) to (i) construct the matrix (ii) diagonalize it to compute the lowest 5 eigenvalues (iii) generate data to plot E vs. k diagram and (iv) generate data to plot the charge density corresponding to the lowest energy eigenvalue at $k = 0$ and lowest two energy eigenvalues at $k = -\frac{\pi}{a}$. Plot the wavefunctions for $-2.5a \leq x \leq 2.5a$. Choose $a=1.0$, no. of k-point (nk)=401.

Choose the G vectors such that $G=0$ is included and there are equal number of G's on +ve and -ve side.

- (3) Take $V_0 = 0.0$ and $a=1.0$. Using 3, 11, 21 and 61 planewaves compute and plot the total energies mentioned in (2). How do the results change? Can you use these to decide the optimal value of the no. of plane waves? If so, what should be a good choice?
- (4) For the case in (3) increase a to 1.5 and 2.5. How do the quantities that you have computed changes? Explain your observation.
- (5) Now consider $V_0 = 0.5$. How many planewaves should you use for this case? Compute and plot the quantities mentioned in Q2. What changes do you observe? Are these in agreement with what you expect from analytical calculation?
- (6) At what value of V_0 perturbation theory breaks down?
- (7) Now keeping V_0 fixed to (i) 0.5 and (ii) 8.5 au, increase a from 1.0 to 10.0 au, in steps of 3.0. Compare how the results change wrt $a=1.0$ and also between (i) and (ii). Rationalize your observations.